PTUT — FINAL YEAR PROJECT

Weekly Progress Report · Week 7 of 12

TaxWise Freelance

Automated Tax Deductions & Invoicing Platform

Mid-Term Progress Report · Live Working Prototype

STUDENT
Muhammad Sohaib Ramzan

ROLL NUMBER
24-ST-004

SUPERVISOR
Dr. Zaigham Mushtaq

DEPARTMENT
Software Engineering Technology

SECTION
A

PROJECT ID
PTUT-PRJ-042

DOMAIN
FinTech, Personal Wealth & Micro-Finance

Source code: github.com/SohaibRamzan01/taxwise-freelance

1. Summary

TaxWise Freelance is a **working prototype**. A freelancer can sign up, scan a receipt, see the deductible amount, send and track invoices, get a quarterly tax estimate, and download a tax-ready PDF report. A full end-to-end test of this journey passes, and the prototype is ready for the live demonstration.

14/14
end-to-end journey steps passing

289
automated tests passing

99%
test coverage on core logic

Task from the plan	Status
End-to-end smoke test of the full user journey	Done — 14 of 14 steps pass (section 4)
Mid-Term Progress Report	Done — this document
Mid-Term Slide Deck	Done — 15 slides with speaker notes
Live demonstration for the supervisor	Ready — demo script in section 8

2. Progress against the 12-week plan

The core product is finished ahead of schedule. Several later milestones (Weeks 8–10) are already built, which leaves time for deployment, real-user testing and the final report.

Week	Milestone	Status
1	Problem discovery & personas	Done
2	Software Requirements Specification	Done
3	Architecture & cloud topology	Done
4	Database, API & wireframes	Done
5	Scaffolding & secure login	Done
6	Tax engine, OCR, invoices, PDF reports	Done
7	**Midterm: live prototype**	**Done — this report**
8	Gemini AI, search, bulk actions, background job	Built early
9	Admin portal, charts, mobile polish	Built early
10	Tests ≥ 70% coverage, security audit	Built early (99%)
11	CI pipeline & cloud deployment	CI done · deployment next

Week	Milestone	Status
12	Final report & viva	Planned

3. What the prototype does

Feature (from the project description)	What works now
Automatically categorise receipts (hardware, internet, travel)	Upload or photograph a receipt → **Tesseract OCR** reads vendor, date and total → category suggested by rules or **Gemini AI** → user confirms
Track tax-deductible expenses	17 categories; deductible = amount × business-use % × category factor (meals 50%)
Invoicing	Clients, line items, INV-0001 numbering, Draft/Sent/Paid/Overdue/Void, invoice PDF; paid invoices become income automatically
Calculate quarterly tax liabilities	Self-employment tax, income tax, QBI, safe harbour, quarterly plan with Paid / Past due / Upcoming status
Tax-ready summary reports	Year or quarter **PDF made with PDFKit**, plus CSV
Extra	Dashboard with charts, search, filters, CSV import/export, bulk actions, notifications, admin portal

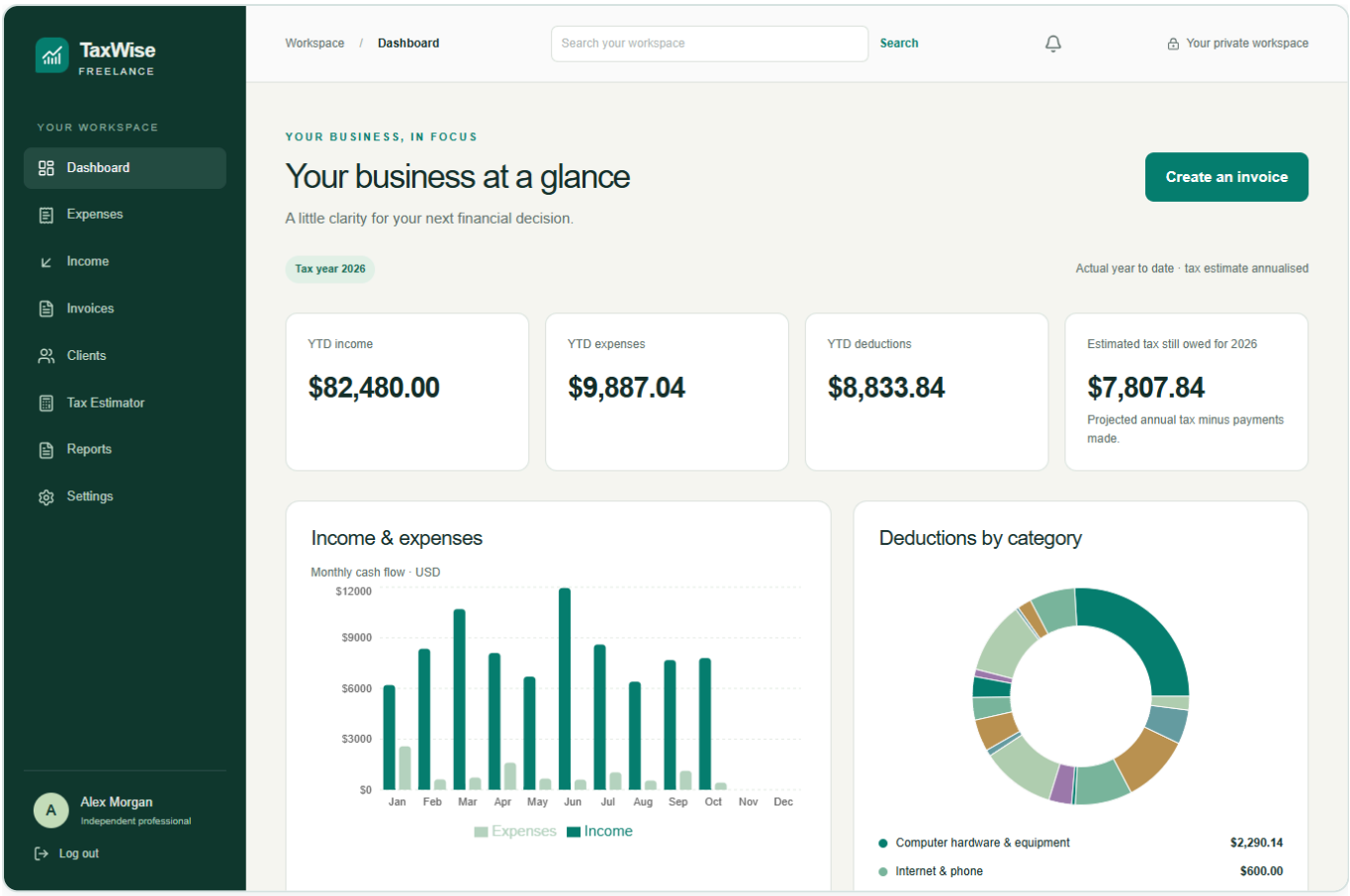


Figure 1 — Dashboard with a full demo year: about $82,000 income and 48 expenses

4. End-to-end smoke test

The smoke test runs the whole journey against the **real running app** (production build, real database, real OCR). It creates a throwaway account and cleans up after itself.

#	Step	Result
1	Health check (app + database)	PASS
2	Sign up a new user	PASS
3	Create a client	PASS
4	Create an invoice	PASS
5	Mark invoice paid → exactly one income entry created	PASS
6	Scan a sample receipt with real Tesseract OCR	PASS
7	Save the reviewed expense and fetch the private receipt image	PASS
8	Tax estimate	PASS
9	Download the tax summary PDF	PASS

#	Step	Result
10	Notifications; normal user blocked from admin pages	PASS
11–13	Clean up expenses, invoices and clients	PASS
14	Log out	PASS

The same test also runs automatically on GitHub Actions on every push, together with lint, type checks, 289 unit tests and a production build.

5. Key numbers

Measure	Value
Automated tests	289 passing
Line coverage on core logic (`src/lib`)	≈ 99% (target 70%)
API endpoints	53 handlers across 36 routes
Screens	13 (login, sign up, dashboard, expenses, income, clients, invoices, invoice print view, tax estimator, tax payments, reports, settings, admin)
Known vulnerabilities in production packages	0
Monthly running cost (planned)	$0

6. Risks and how they are handled

Risk	Plan
OCR misreads blurry receipts	User always reviews scanned details; manual entry always available
Free cloud tiers have limits (upload size, cold starts)	Phone photos are shrunk before upload; Neon pooled connection; health check
Tax rules change each year	Rules are data files per year; the engine is tested with hand-checked numbers
User profiles not yet confirmed with real freelancers	Short testing sessions with real freelancers planned for Weeks 8–9

7. Plan for the rest of the semester

- **Week 8:** deploy to Vercel + Neon with SSL; run the smoke test against the live URL.
- **Weeks 8–9:** test with real freelancers and apply supervisor feedback from the midterm.
- **Weeks 10–11:** final security review, performance check against the latency target, production health checks.
- **Week 12:** final FYP report, final slides and viva preparation.

8. Live demo script (about 6 minutes)

Before the demo: start the database and the app, open the browser at the login page, and keep `public/samples/hardware.png` ready.

1. **Log in** as the demo user (`demo@taxwise.example`). Point out the dashboard: income, expenses, deductions, tax still owed, and the monthly chart.
2. **Expenses → Scan receipt.** Upload `hardware.png`. Show that OCR filled in the vendor, date and total, and that the category is *Computer hardware & equipment* with an "auto" badge. Click **Save**.
3. **Show the deductible column.** Point at the new row, then at *PTCL $80 at 75% → $60* and a meal counted at 50%.
4. **Invoices.** Find **INV-0014 (Northline Studio, Sent)** and click **Mark paid**. Open **Income** to show the new income entry was created automatically.
5. **Tax Estimator.** Show the line-by-line calculation, the quarterly plan (Q1–Q2 Paid, Q3 Past due $150, Q4 Upcoming) and the two figures: *tax still owed* vs *payment target still due* (safe harbour).
6. **Reports.** Choose 2026 and click **Download PDF** — open the tax-ready summary.
7. **(Optional) Admin.** Log out and log in as `admin@taxwise.example` to show the analytics charts and audit log; a normal user gets "forbidden".

If something fails: run `npm run smoke -- http://localhost:3000` — it shows the same journey passing step by step.